Active Directory Home Lab – Windows Server 2025

Project Overview

This project demonstrates the deployment and administration of an ***Active Directory environment*** using ***Windows Server 2025*** and a ***Windows 11 client workstation***. The lab simulates common enterprise IT support and system administration tasks such as domain controller deployment, DNS configuration, user lifecycle management, domain authentication, and role-based access control.

The goal of this lab is to practice and demonstrate the practical skills required for roles such as:

- IT Support
- IT Technician
- System Administrator
- Infrastructure Engineer
- SOC Analyst (identity infrastructure knowledge)

The lab environment replicates a small company network with departmental structure, access control, and centralized authentication.

Lab Architecture

Network Design:

- pfSense Firewall – 192.168.10.1
- Windows 11 Client – 192.168.10.10
- Domain Controller (DC01) – 192.168.10.20
- Ubuntu Server – 192.168.10.30
- Ubuntu Desktop – 192.168.10.40

Domain Name:
lab.local

The Domain Controller (DC01) hosts the following services:

- Active Directory Domain Services
- DNS Server

All domain devices use the Domain Controller as their primary DNS server.

Virtual Machine Inventory

Machine	Role	IP Address
pfSense	Firewall / Gateway	192.168.10.1
DC01	Domain Controller	192.168.10.20
PC01	Windows 11 Client	192.168.10.10
Ubuntu Server	SIEM / Wazuh (future SOC projects)	192.168.10.30
Ubuntu Desktop	Linux workstation	192.168.10.40

Technologies Used

- Windows Server 2025
- Active Directory Domain Services (AD DS)

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- DNS
- Windows 11
- pfSense Firewall
- UTM Virtualization
- Organizational Units (OU)
- Security Groups
- NTFS Permissions
- File Sharing

Active Directory Deployment

Domain Controller configuration:

- Hostname: DC01
- Domain: lab.local
- IP Address: 192.168.10.20
- DNS Server: 192.168.10.20

Active Directory Domain Services (AD DS) was installed, and the server was promoted to a Domain Controller.

The server now provides:

- Domain authentication
- Centralized identity management
- DNS name resolution

Organizational Unit Structure

To simulate a corporate structure, Organizational Units were created for each department.

lab.local

- IT
- HR
- Finance
- Sales
- Disabled_Users

This structure allows departments to be managed independently and enables targeted administration and policies.

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Security Groups

Security groups were created to manage access control based on department roles.

IT_Admins
HR_Users
Finance_Users
Sales_Users

Using security groups instead of assigning permissions directly to users simplifies administration and improves scalability.

User Management

Users were created within their respective departmental OUs.

Example user structure:

IT

1. Daniel Carter
2. Sofia Nilsson
3. Marcus Lee
4. Olivia Berg

HR

1. Emma Johnson
2. Lucas Andersson
3. Nora Patel
4. Ethan Walker

Finance

1. Wei Chen
2. Mia Eriksson
3. Alexander Brown
4. Lina Svensson

Sales

1. Noah Garcia
2. Ella Johansson
3. Samuel Kim
4. Ava Thompson

Each user was assigned to their respective department security group.

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Domain Join

The Windows 11 workstation (PC01) was joined to the Active Directory domain.

Domain:
lab.local

Users can now authenticate using domain credentials:
LAB\username

Authentication is handled by the Domain Controller.

User Lifecycle Simulation

This lab demonstrates several common IT support tasks.

User Onboarding

- Create new user accounts
- Assign users to security groups
- Configure initial passwords

Password Reset

Password reset simulated using Active Directory Users and Computers.

User Offboarding

Offboarding was simulated by:

- Disabling the user account
- Moving the account to the Disabled_Users OU

This prevents former employees from accessing network resources.

File Share Access Control

A shared folder was created on the Domain Controller.

Folder path:
C:\Finance_Share

Share permissions were configured using security groups.
Permissions:
Finance_Users – Full Control

Only members of the Finance department are allowed to access the shared resource.

Access Control Testing

Testing confirmed that permissions were correctly enforced.
Finance User
LAB\wchen

Access result:
Access granted to:
\DC01\Finance_Share

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Non-Finance User
Example:
LAB\dcarter

Access result:
Access denied

This confirms that role-based access control is functioning correctly.

Key Skills Demonstrated

- Active Directory deployment
- DNS configuration
- Organizational Unit design
- Security group management
- User lifecycle management
- Domain authentication
- File share permission management
- Role based access control
- Enterprise network architecture simulation

Screenshots

The following screenshots document the entire deployment and administration process.

1. Server Manager after Windows Server installation
2. Static IP configuration on DC01
3. Active Directory Domain Services role installation
4. Promote server to Domain Controller
5. Create new forest (lab.local)
6. Domain Controller options configuration
7. NetBIOS domain name configuration
8. Active Directory Users and Computers console
9. Organizational Unit structure created
10. Department security groups created
11. First user account created
12. User added to security group
13. Department users created in OUs
14. Windows 11 joined to domain
15. Domain login screen
16. Domain user successfully logged into workstation
17. Password reset operation
18. User account disabled and moved to Disabled_Users OU
19. Shared folder permissions configuration
20. File share access test from Windows 11

Screenshots are stored in the repository under: /screenshots

Lessons Learned

This project reinforced several key concepts about enterprise identity infrastructure:

- Active Directory relies heavily on DNS
- Organizational Units provide administrative structure
- Security groups simplify permission management
- Domain controllers centralize authentication
- Access control should always be group-based rather than user-based

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These concepts are fundamental in real enterprise IT environments.

Conclusion

This lab demonstrates the successful deployment and administration of a fully functional Active Directory environment.

The project simulates common tasks performed by IT support and system administrators including identity management, authentication services, and access control.

The lab environment will be expanded in future projects to include:

1. Wazuh SIEM deployment
2. Security monitoring
3. Sysmon log analysis
4. Threat detection
5. SOC incident simulations

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Future Improvements

1. Wazuh SIEM deployment
2. Sysmon log monitoring
3. Threat detection
4. SOC automation
5. Security incident simulations

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